

DURANT, OK, USA

WMO: 742513

Lat: 33.942N

Lon: 96.394W

Elev: 213

StdP: 98.79

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53937

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.3	(c) -3.8	(d) -12.5	(e) 1.3	(f) -2.5	(g) -10.2	(h) 1.6	(i) 0.6	(j) 11.6	(k) 9.6	(l) 10.7	(m) 11.1	(n) 3.6	(o) 340	(p) 0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	38.0	24.5	37.2	24.4	36.0	24.2	26.7	33.5	25.9	32.9	25.5	32.6	3.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.6	29.7	24.0	19.4	28.6	23.7	19.0	28.4	84.6	32.9	81.8	32.7	79.1	32.9	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	9.0	8.2	-10.2	39.9	1.9	2.1	-11.6	41.3	-12.7	42.6	-13.8	43.7	-15.2	45.2
			-11.0	27.5	1.7	0.8	-12.2	28.1	-13.2	28.5	-14.1	29.0	-15.3	29.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	6.7	8.2	13.5	17.4	22.1	27.0	28.8	28.8	25.0	18.9	12.9	7.8	
	DBStd	9.01	5.41	5.87	5.08	4.37	3.73	2.42	2.46	2.71	3.24	4.43	4.85	5.04	
	HDD10.0	384	131	95	24	4	0	0	0	0	0	1	25	104	
	HDD18.3	1437	362	287	162	66	12	0	0	0	1	48	171	329	
	CDD10.0	3357	28	44	133	227	377	510	582	581	450	277	113	34	
	CDD18.3	1366	1	2	12	39	130	260	324	323	200	66	9	1	
	CDH23.3	15059	4	17	92	323	1161	2852	3962	3943	2015	619	65	7	
	CDH26.7	7478	0	2	11	61	409	1416	2198	2209	962	202	9	0	
Wind	WSAvg	3.6	3.8	4.1	4.4	4.4	3.9	3.5	2.9	2.7	2.8	3.2	3.7	3.7	
Precipitation	PrecAvg	1114	53	70	98	107	138	127	70	74	130	106	85	71	
	PrecMax	1523	172	183	250	323	319	318	221	240	291	452	255	208	
	PrecMin	540	0	2	25	7	14	40	0	0	6	4	0	7	
	PrecStd	242	40	41	55	77	81	76	59	54	84	86	58	53	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.6	26.0	27.8	31.0	33.9	37.6	40.1	41.2	37.7	32.7	27.7	23.9	
		MCWB	14.9	16.9	18.5	20.8	23.1	25.1	24.7	24.5	22.8	21.9	18.7	16.7	
	2%	DB	21.2	22.5	26.1	27.9	32.4	36.1	38.0	38.8	36.0	31.1	25.2	21.2	
		MCWB	14.6	15.7	18.0	19.8	23.1	24.0	24.6	24.5	22.9	21.7	17.7	15.0	
	5%	DB	17.9	20.2	23.7	27.1	31.1	34.8	37.2	37.4	33.7	28.9	22.7	18.7	
		MCWB	13.3	15.0	17.1	19.4	22.5	24.0	24.5	24.5	23.0	20.0	16.8	14.4	
	10%	DB	16.2	17.7	22.2	25.2	28.8	32.8	35.9	36.0	32.3	27.2	21.2	17.0	
		MCWB	11.0	13.6	16.1	18.4	22.2	24.3	24.5	24.4	22.7	19.4	16.4	12.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	19.1	20.7	23.0	25.7	27.1	27.8	27.7	26.0	24.4	20.8	19.3
			MCDB	21.1	23.0	25.0	27.7	31.2	33.7	34.3	34.4	32.1	28.7	23.7	22.5
2%		WB	17.1	17.8	19.5	22.0	24.7	26.1	26.6	26.6	25.1	23.4	19.7	17.5	
		MCDB	18.7	20.2	23.7	26.4	29.9	32.9	33.6	33.4	31.2	28.2	23.2	19.4	
5%		WB	14.4	16.2	18.6	21.0	23.9	25.5	25.8	25.8	24.4	22.4	18.6	15.6	
		MCDB	17.7	19.2	22.4	24.8	28.5	32.1	33.0	33.1	30.2	26.8	21.8	17.7	
10%		WB	11.7	14.3	17.4	19.9	23.1	25.0	25.4	25.3	23.7	21.2	17.2	12.9	
		MCDB	15.2	17.6	20.6	23.8	27.5	31.3	32.7	33.0	29.3	24.7	19.9	15.5	
Mean Daily Temperature Range	5% DB	MDBR	11.4	11.2	11.9	12.1	11.1	11.1	11.8	12.2	12.2	12.7	12.0	11.2	
		MCDBR	15.2	14.5	13.8	13.3	12.7	12.8	14.0	14.0	14.2	14.4	13.8	14.7	
		MCWBR	9.7	9.2	7.7	7.0	5.3	4.0	3.6	3.6	4.5	6.4	8.0	9.9	
	5% WB	MCDBR	12.9	12.8	11.4	11.0	10.6	10.9	11.6	11.9	11.1	11.2	11.5	12.1	
		MCWBR	10.0	9.4	7.2	7.4	5.5	4.6	4.2	4.1	4.4	6.1	8.6	10.3	
Clear-Sky Solar Irradiance	taub		0.310	0.319	0.354	0.385	0.422	0.432	0.438	0.442	0.395	0.352	0.332	0.311	
	taud		2.491	2.483	2.405	2.333	2.295	2.304	2.325	2.295	2.375	2.483	2.480	2.492	
	Ebn at Noon		905	933	920	902	868	855	848	839	866	883	869	880	
	Edh at Noon		85	95	112	126	132	131	128	129	114	95	86	81	
All-Sky Solar Radiation	RadAvg		2.73	3.24	4.31	5.25	5.69	6.46	6.60	6.05	5.05	3.93	2.92	2.38	
	RadStd		0.28	0.48	0.32	0.40	0.56	0.69	0.39	0.39	0.49	0.57	0.37	0.30	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page